

● MEDIUM

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

06

10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS

A company fills boxes with approximately 23 pounds of oranges. To test the accuracy of the filling process, 344 boxes of oranges were selected at random and weighed. Based on the sample, it is estimated that the average weight of all boxes of oranges filled by the company in an 8-hour period is 23.1 pounds, with an associated margin of error of 0.19 pounds. Which of the following is the best interpretation of this estimate?

- A. Plausible values for the average weight of all boxes of oranges filled by the company are between 22.91 pounds and 23.29 pounds.
- B. Plausible values for the average weight of all boxes of oranges filled by the company are less than 22.91 pounds or greater than 23.29 pounds.
- C. The average weight of all boxes of oranges filled by the company is less than 23.01 pounds.
- D. The average weight of all boxes of oranges filled by the company is greater than 23.01 pounds.

A number n is increased 6%. If the result is 318, what is the value of n ?

- A. 199
- B. 299
- C. 300
- D. 337

If one of these tiles is selected at random, what is the probability of selecting a red tile? (Express your answer as a decimal or fraction, not as a percent.)

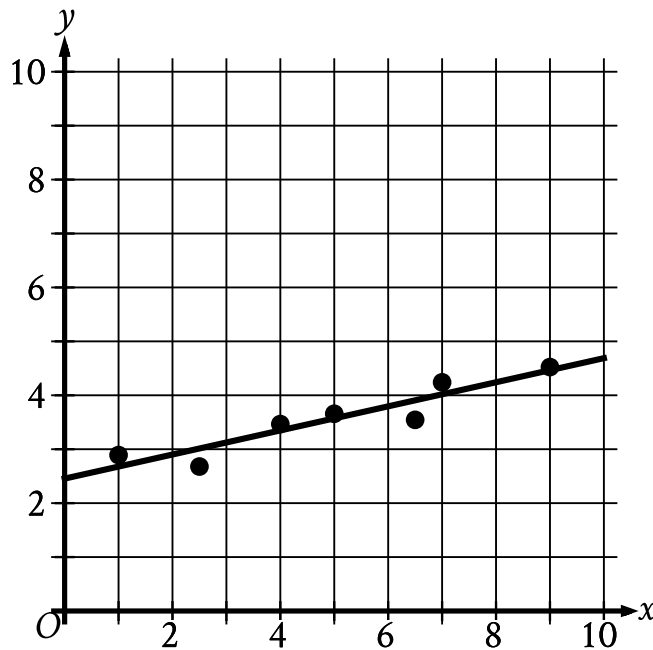
	Red	Blue	Yellow	Total
Square	10	20	25	55
Pentagon	20	10	15	45
Total	30	30	40	100

If one of these tiles is selected at random, what is the probability of selecting a red tile? (Express your answer as a decimal or fraction, not as a percent.)

.	/	.	/	.
0	0	0	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7	7	7	7	7
8	8	8	8	8
9	9	9	9	9

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In the given scatterplot, a line of best fit for the data is shown.



- The scatterplot has 7 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
 - The line of best fit slants up from left to right.
 - The line of best fit passes through the following approximate coordinates:
 - (0 comma 2.5)
 - (5 comma 3.6)
 - (10 comma 4.7)

Which of the following is closest to the slope of the line of best fit?

- A. -2.45
- B. -0.22

C. 0.22

D. 2.45

Ages of 20 Students Enrolled in a College Class

Age	Frequency
18	6
19	5
20	4
21	2
22	1
23	1
30	1

The table above shows the distribution of ages of the 20 students enrolled in a college class. Which of the following gives the correct order of the mean, median, and mode of the ages?

- A. mode < median < mean
- B. mode < mean < median
- C. median < mode < mean
- D. mean < mode < median

Q6

PROBLEM-SOLVING AND DATA ANALYSIS

An object has a mass of 168 grams and a volume of 24 cubic centimeters. What is the density, in grams per cubic centimeter, of the object?

- A. 7
- B. 144
- C. 192
- D. 4,032

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

Based on a random sample from a population, a researcher estimated that the mean value of a certain variable for the population is 20.5, with an associated margin of error of 1. Which of the following is the most appropriate conclusion?

- A. It is plausible that the actual mean value of the variable for the population is between 19.5 and 21.5.
- B. It is not possible that the mean value of the variable for the population is less than 19.5 or greater than 21.5.
- C. Every value of the variable in the population is between 19.5 and 21.5.
- D. The mean value of the variable for the population is 20.5.

Q8

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

The amount of Hanna's bill for a food order was \$ 50. Hanna gave a tip of 20% of the amount of the bill. What is the amount, in dollars, of the tip Hanna gave?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A box contains 13 red pens and 37 blue pens. If one of these pens is selected at random, what is the probability of selecting a red pen? (Express your answer as a decimal or fraction, not as a percent.)

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Each year, the value of an investment increases by 0.49% of its value the previous year. Which of the following functions best models how the value of the investment changes over time?

- A. Decreasing exponential
- B. Decreasing linear
- C. Increasing exponential
- D. Increasing linear